Name: Mandar Sanju Bavdane **CWID: 20041513**

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Results

The table below shows the timing measurements for each of the twelve experiments. nmin is the smallest array size that took at least 30 ms and nmax is the largest array that could be sorted within the available time budget.

Experiment	nmin	tmin_ms	nmax	tmax_ms	tmax/tmin
SC	6000	38	450000	204135	5371.97368
SS	6000	37	450000	203376	5496.64865
SR	6000	37	450000	223977	6053.43243
IC	9500000	33	100000000	310	9.39393939
IS	10000000	33	100000000	400	12.1212121
IR	7000	38	1000000	159807	4205.44737
MC	700000	37	100000000	7126	192.594595
MS	700000	36	100000000	7813	217.027778
MR	300000	36	100000000	13550	376.388889
QC	6500	33	43399	1484	44.969697
QS	400000	34	100000000	519775	15287.5
QR	200000	28	100000000	176499	6303.53571

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Analysis

For each experiment, the three theoretical ratios (nmax)/(nmin) are computed for:
f1(n)=n (Linear) , f2(n)=n ln n (n log n) and f3(n)=n2 (Quadratic)

The measured ratio tmax/tmin is compared to these theoretical ratios to identify the empirical complexity.

Experiment	tmax/tmin	n ratio	n log n ratio	n^2 ratio	Behaviour
SC	5371.97368	75	112.2218012	5625	Quadratic
SS	5496.64865	75	112.2218012	5625	Quadratic
SR	6053.43243	75	112.2218012	5625	Quadratic
IC	9.39393939	10.52631579	12.06848122	110.8033	Linear
IS	12.1212121	10	11.42857143	100	Linear
IR	4205.44737	142.8571429	222.9183361	20408.16	Quadratic
MC	192.594595	142.8571429	195.5240331	20408.16	n log n
MS	217.027778	142.8571429	195.5240331	20408.16	n log n
MR	376.388889	333.3333333	486.8737687	111111.1	n log n
QC	44.969697	6.676769231	8.120662345	44.57925	Quadratic
QS	15287.5	250	357.0115285	62500	n log n
QR	6303.53571	500	754.5703388	250000	n log n

Per Experiment Analysis

SelectionSort on Constant Input (SC)

SelectionSort is theoretically $\theta(n^2)$ in all cases as it always makes n(n-1)/2 comparisons regardless of the input. The measured ratio of 5,372 is very close to the theoretical n^2 ratio of 5,625 which confirms Quadratic behaviour.

SelectionSort on Sorted Input (SS)

Similar theoretical expectation as of SC: $\theta(n^2)$ unconditionally. The measured ratio of 5,497 is again extremely close to the n^2 ratio of 5,625 which confirms Quadratic behaviour. SelectionSort makes the same number of comparisons regardless of whether the array is already sorted.

SelectionSort on Random Input (SR)

Theoretical expectation: $\theta(n^2)$. The measured ratio of 6,053 which slightly exceeds the n^2 ratio of 5,625. This small overshoot is within the noise expected from memory access patterns and cache behaviour on larger random arrays, and the overall behaviour is clearly Quadratic, as expected.

InsertionSort on Constant Input (IC)

For a constant array, InsertionSort, scans each element and finds that it is already in place on the very first comparison, resulting in best-case $\theta(n)$ behaviour. The measured ratio of 9 is very close to the n ratio of 11 confirming Linear behaviour. This matches the best-case W(n) theory exactly.

InsertionSort on Sorted Input (IS)

Theoretically a sorted array is the best case for InsertionSort as each element needs only one comparison. The measured ratio of 12 falls between the n ratio (10) and the n ln n ratio (11). It is closest to n ln n in relative terms. The slight deviation from pure linear may reflect function call overhead and loop control costs that scale as O(n) with a larger constant than IC. Overall the behaviour is very close to linear which is consistent with best-case theory.

InsertionSort on Random Input (IR)

For random input, InsertionSort exhibits its average-case $\theta(n^2)$ behaviour, since on average each element must be shifted past half the previously sorted elements. The measured ratio of 4,205 is substantially larger than the n ratio (143) and the n ln n ratio (223) but finally falls closer to Quadratic. The ratio does not reach the full n^2 prediction, likely because the constant factor of the quadratic term is less than 1 for this implementation. Behaviour is clearly Quadratic.

MergeSort on Constant Input (MC)

MergeSort is **θ** (n log n) in all cases. For constant input the measured ratio of 193 lies between the n ratio (143) and the n ln n ratio (196), and is extremely close to the n ln n ratio. This confirms n log n behaviour which perfectly matches the theory.

MergeSort on Sorted Input (MS)

Again MergeSort is **θ** (n log n) regardless of input. The measured ratio of 217 is close to the n ln n ratio of 196. The slightly higher measured ratio compared to MC may stem from cache effects when merging a sorted array, but the behaviour is clearly n log n, matching theoretical expectation.

MergeSort on Random Input (MR)

MergeSort is **θ** (n log n) theoretically. The measured ratio of 376 is closest to the n ratio of 333 versus the n ln n ratio of 487. While the ratio falls between n and n ln n, this is likely due to practical effects: the random input induces worse cache behaviour during merging, reducing the effective constant factor compared to the theoretical n ln n prediction.

QuickSort on Constant Input (QC)

When all elements are equal, a randomly chosen pivot always partitions the array into one empty subarray and one subarray of size n-1, regardless of the pivot position. This produces worst-case $O(n^2)$ behaviour. The measured ratio of 45 matches the n^2 ratio of 45 essentially exactly, confirming the Quadratic behaviour.

QuickSort on Sorted Input (QS)

Although QuickSort uses a random pivot, which should give average-case **θ** (n log n) even on sorted input, the measured ratio of 15,288 far exceeds both the n ratio (250) and n ln n ratio (357), and is closer to the n^2 ratio of 62,500. The behaviour is labelled Quadratic because the measured value is vastly closer to the n^2 regime than to n ln n. A likely explanation is that the random pivot still produces poor partitions more often on sorted input than on truly random input or that the nmax used (100 million) which stresses the stack depth and memory system in ways that inflate the constant factor substantially.

QuickSort on Random Input (QR)

For random input, QuickSort with a random pivot achieves its average case **θ** (n log n) behaviour. The measured ratio of 6,304 is larger than the n ln n ratio of 755 but is by far the closest of the three theoretical values in relative terms:

(measured/n^2 = 2.5%, measured/n log n = 835%)

The behaviour is n log n, as expected for the average case. The higher-than-predicted absolute ratio is consistent with larger constant factors from cache misses during random access pivoting on very large arrays.